

SHASHIKA FERNANDO

Software Engineering

<https://github.com/Shashika616>

shashikauda@gmail.com | +94 77 947 3203 | Colombo | <https://www.linkedin.com/in/shashika-fernando-5399062b5/>

PROFESSIONAL SUMMARY

Full-Stack Software Engineer specializing in Identity & Access Management (IAM) and cloud-native architecture, with experience in designing secure, scalable enterprise solutions. Proficient in authentication/authorization systems, microservices architecture, and REST API development. Delivered measurable impact including 40% improvement in team productivity, 35% faster issue resolution, and 99.5%+ system uptime in production environments. Seeking opportunities to contribute expertise in IAM and full-stack development within secure, engineering teams.

EDUCATION

Bachelor of Computer Science (Honours) | 2021 – 2026

University of Ruhuna, Matara | GPA : 3.87/4.0

G. C. E Advanced Level Examination | 2019

Ananda College, Colombo 10 | Physical Science Stream

PROFESSIONAL EXPERIENCE

Internship: Software Engineer | WSO2 LLC | 2024

- Implemented single sign-on (SSO) solutions using WSO2 Identity Server and Asgardeo, configuring SAML 2.0 and OpenID Connect (OIDC) protocols for enterprise application authentication. Supported multi-tenant identity governance with role-based access control (RBAC), enabling secure federated identity across distributed systems serving 1000+ enterprise users globally.
- Designed and deployed a ServiceNow portal for engineer feedback management with comprehensive access control layers (RBAC and ACLs). Built REST APIs, automated notification workflows, and interactive dashboards, reducing manual operational overhead by 40% and improving team productivity.
- Collaborated with cross-functional teams (Product, QA, DevOps, Security) in an agile environment, participating in sprint planning, code reviews, and deployment cycles for production systems. Contributed to best practices documentation and mentored junior team members on IAM concepts and secure coding principles.

Software Engineer (Part-time) | Seefa Business Solutions | 2026– Present

- Engineered a customer management application to track organizational accounts, commissions, payments, and promotion schemes. Implemented secure user authentication, fine-grained permission controls, and audit logging for financial transaction transparency. Delivered RESTful APIs supporting 50+ concurrent users with 99.5% uptime.
- Built an end-to-end ticket lifecycle management system for product change requests and new feature implementations. Implemented automated routing, escalation workflows, and role-based ticket assignment, improving resolution time by 35% and enhancing customer satisfaction.

PROJECTS & RESEARCH

Novel Password Hashing Pipeline | 2025 – Present

- Engineered a secure password transformation layer integrated with Argon2 hashing to strengthen entropy for weak passwords by over 7x. Implemented deterministic authentication achieving 100% success rate with 250–500ms latency, significantly enhancing resistance to brute-force attacks while maintaining real-time usability.
- Incorporated security best practices and cryptographic standards (NIST guidelines), publishing technical documentation for team knowledge transfer and future system maintenance.
- Technologies: Python.

Tele-communication application for a Distributed Environment | 2025

- Architected and deployed production-ready microservices using Node.js/Express with stateless design patterns and REST API standards. Implemented asynchronous message queue system (RabbitMQ) enabling fault-tolerant inter-service communication and decoupled system components.
- Implemented persistent data layer with MongoDB and high-performance caching layer using Redis, achieving 99% uptime SLA and sub-100ms response times across 50+ API endpoints handling 1000+ concurrent connections.
- Developed automated deployment pipeline and conducted load testing to validate system performance under peak traffic conditions. Containerized services using Docker for consistent development, staging, and production environments.
- Technologies: Flutter, Nodejs, Express.js, MongoDB, Redis, RabbitMQ, Docker.

Rupavahini GatePass & Visitor Management System | 2024

- Delivered a full-stack MERN-based identity and access management solution for physical facility visitor tracking. Implemented secure digital gate pass generation with authentication validation, role-based user permissions, and comprehensive audit trails for compliance.
- Digitized entry processes, reducing paper usage by 90% and ensuring compliance with organizational security policies. System processes 500+ visitor entries monthly with zero security incidents.
- Designed responsive UI/UX with Figma prototypes and managed project workflow using Trello, demonstrating ability to bridge design and development across cross-functional teams.
- Technologies: React, Nodejs, Express.js, MongoDB, Figma, Trello.

Skills

- **IAM Platforms & Protocols:** WSO2 Identity Server, Asgardeo, SAML 2.0, OAuth 2.0, OpenID Connect (OIDC), SCIM, MFA, Password-less Authentication, User Lifecycle Management, RBAC, ACLs
- **Cloud & Identity Services:** Azure fundamentals
- **Programming:** JAVA, C, C#, JavaScript, Python, HTML | **Database:** MySQL, MongoDB, PostgreSQL
- **Technical Fields:** Research and Development, Cyber Security, Web Development, AI and Machine learning
- **Frameworks/Tools:** Springboot, Node, .NET, GIT, Google Colab, Redis, Servicenow
- **Soft Skills:** Technical Documentation, Problem-Solving, Team Collaboration, Adaptability, Continuous Learning

Certificates

- **WSO2 Certified Associate – Identity & Access Management (CID-04598722) – [View Certificate](#)**
- **UI/UX – [View Certificate](#)**
- **Cloud Computing – [View Certificate](#)**
- **React JS – [View Certificate](#)**
- **JavaScript Development – [View Certificate](#)**

References

Mr. L.L. Gihan Chathuranga Lecturer (Probationary)
University of Ruhuna
Phone : +94 77 659 2307
Email : gihan@dcs.ruh.ac.lk

Dr. K. D. C. G Kapugama Senior Lecturer (Grade II)
University of Ruhuna
Phone : +94 75 081 2584
Email : charaka@dcs.ruh.ac.lk